FPGA/GPU inspection benchmarks

Final campaign: 19 October 2026 · Report: 20 October 2026

This report compares five implemented inspection paths on the same host: Buffering, Streaming, Resource-optimized streaming, GPU_HOST, and GPU_P2P. The GPU paths use the original FP32 model; FPGA paths use the quantized W8A8 model. These are system comparisons, not equal-precision kernel comparisons.

Main result. Resource-optimized streaming reduces routed wrapper BRAM by 83.95% and DSPs by 51.20% versus Streaming. Its resident latency at 4 MiB is 1.119 ms, 1.42% above Streaming. It is 2.67× faster than GPU_HOST at 4 MiB and 1.32× faster at the largest frame.

Important exception. At exactly 4,194,240 bytes, Resource-optimized streaming takes **16.824 ms** versus **2.990 ms** for GPU_HOST: the GPU is 5.63× faster. Adding one byte lowers FPGA latency to about 1.119 ms. The recorded FPGA preprocessing interval accounts for 16.777 ms at the slow boundary. This result prevents a blanket claim that the FPGA system is always faster.

Variant	4 MiB resident median / p95 (ms)	4 MiB file-inclusive median / p95 (ms)
Buffering	1.431 / 1.432	2.556 / 3.071
Streaming	1.104 / 1.105	2.256 / 2.645
Resource-optimized streaming	1.119 / 1.121	2.243 / 2.460
GPU_HOST	2.985 / 3.067	4.185 / 4.372
GPU_P2P	3.771 / 4.089	5.007 / 5.267

P2P outcome. GPU_P2P is slower than GPU_HOST in this setup. At the largest frame, its resident median is 41.851 ms versus 13.917 ms for GPU_HOST. The measured P2P forwarding interval is 32.682 ms. Autonomous polling passed board admission; that establishes correctness, not a latency advantage.

Evidence: 30,000 primary measured calls, 6,000 closing-baseline calls, and 600 separate GPU profiling calls. All accepted baseline drift is below 5%. Two file-inclusive cases were replaced by a predefined matched recheck; original evidence is preserved.

Reading guide: pages 3–7 explain GPU timing; pages 8–10 compare FPGA variants; pages 11–12 compare Resource-optimized streaming with GPU_HOST. Resources, validation, and complete tables follow.

Measurement protocol and interpretation

Report label	Iteration / implementation
Buffering	OPT_0_NONSTREAMING; buffers all 65,536 pixels before CNN consumption
Streaming	OPT_1_STREAMING; original CNN with streaming preprocessing
Resource-optimized streaming	OPT_6_BRAM_FIFO1024_DSP_B2_RF72_RF288; native FIFO profiling capacity 1024; conv3/4/5/6 reuse 72/72/288/288
GPU_HOST	OPT_7_GPU_HOST; host sends input to GPU and FPGA; host relays GPU result
GPU_P2P	OPT_8_GPU_P2P; FPGA forwards input to GPU and autonomously polls result mailbox

Machine: hacc-gpu-u55c-01; U55C device2, PCIe 0000:81:00.0, serial XFL13BA4HVZYA; MI210 GPU0, PCIe 0000:03:00.0. Host thread pinned to CPU64/NUMA1; GPU is on NUMA0. FPGA link: PCIe Gen3 ×16. All comparisons use the same devices and 128 MiB frame allocation.

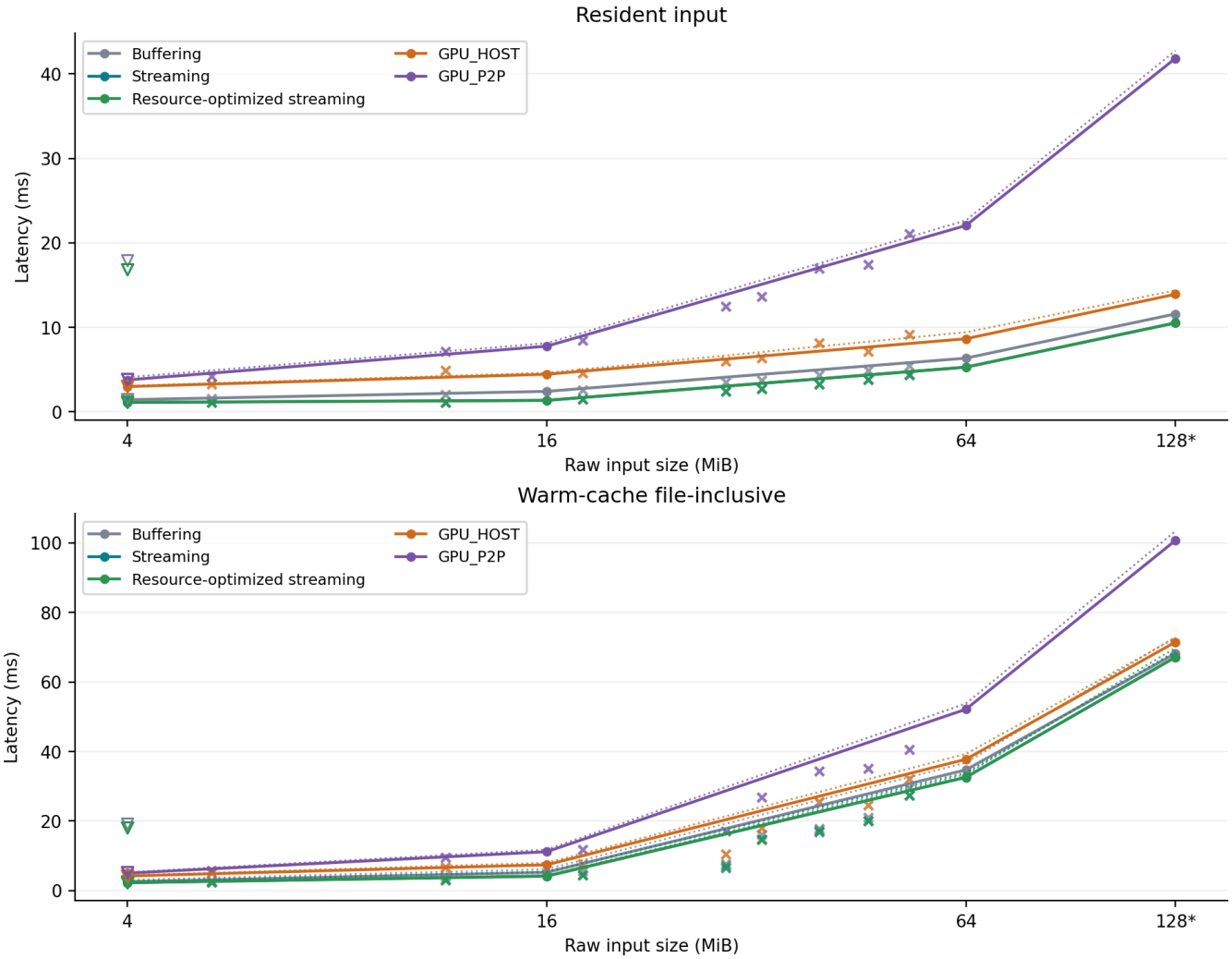
Workloads: eight real bitstreams, four synthetic sizes, and three preprocessing boundary inputs. Each variant has 20 warmups and 200 measured trials per workload per scope. Each image passes 520 board-admission calls and a pilot. The sequence is Streaming → Buffering → Resource-optimized streaming → GPU_HOST → GPU_P2P → Streaming repeat.

Resident: prepared input invocation to host-observed verdict; excludes file reading, preparation, initial allocation, programming, and diagnostic CSR readback. **File-inclusive:** file open/read through preparation/copy to host-observed verdict, with a warm filesystem cache. Every call transfers the entire raw input.

Statistics: solid lines show medians; dotted lines show p95 unless stated otherwise. Synthetic points are connected; real inputs use crosses and boundary cases use hollow triangles. The final synthetic point is 134,217,664 bytes (128 MiB minus a 64-byte header). Breakdown stacks show means of consecutive intervals, not sums of medians.

Interpretation limits: streaming intervals overlap. FPGA CNN first-input-to-score includes input stalls. GPU diagnostic traces add profiling and explicit synchronization; they are not interchangeable with unprofiled E2E. Samples quantify within-run variation; no independent-day confidence interval or deployment-level accuracy equivalence is claimed.

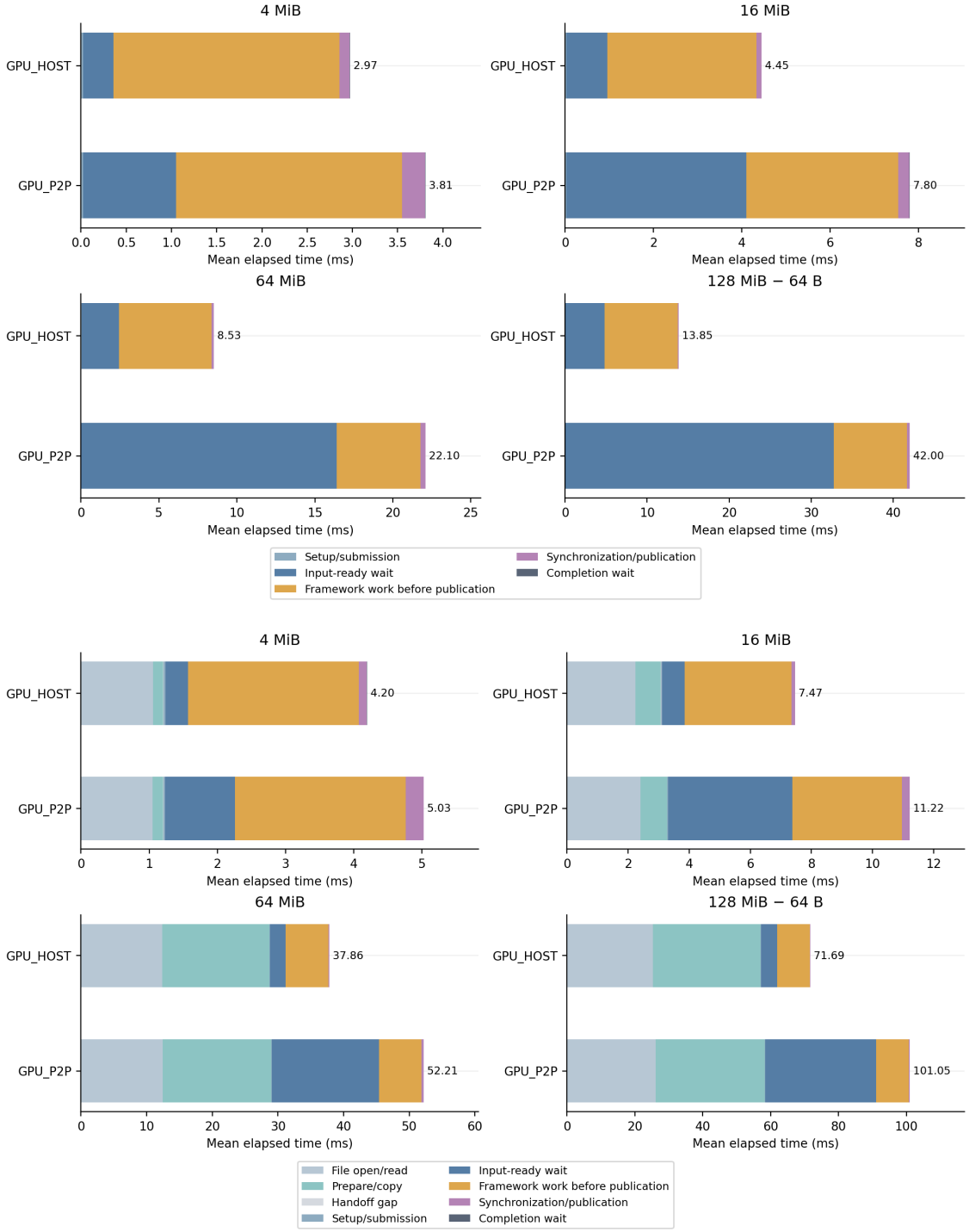
Latency across the complete workload range



The FPGA paths have the lowest resident latencies for the real and synthetic workloads, but not for the slow 4,194,240-byte boundary case (hollow triangles). Buffering exposes the CNN interval after preprocessing; Streaming overlaps these activities. Resource-optimized streaming remains close to Streaming while reducing resources. File-inclusive costs increasingly reflect host reading and copying as input size grows.

Real inputs are separate workloads, not interpolation checkpoints for the synthetic curve. The three nearly coincident boundary sizes are expanded on page 14; their p95 values are in the full tables. Dotted p95 curves apply to synthetic inputs and describe tail latency, not confidence bands.

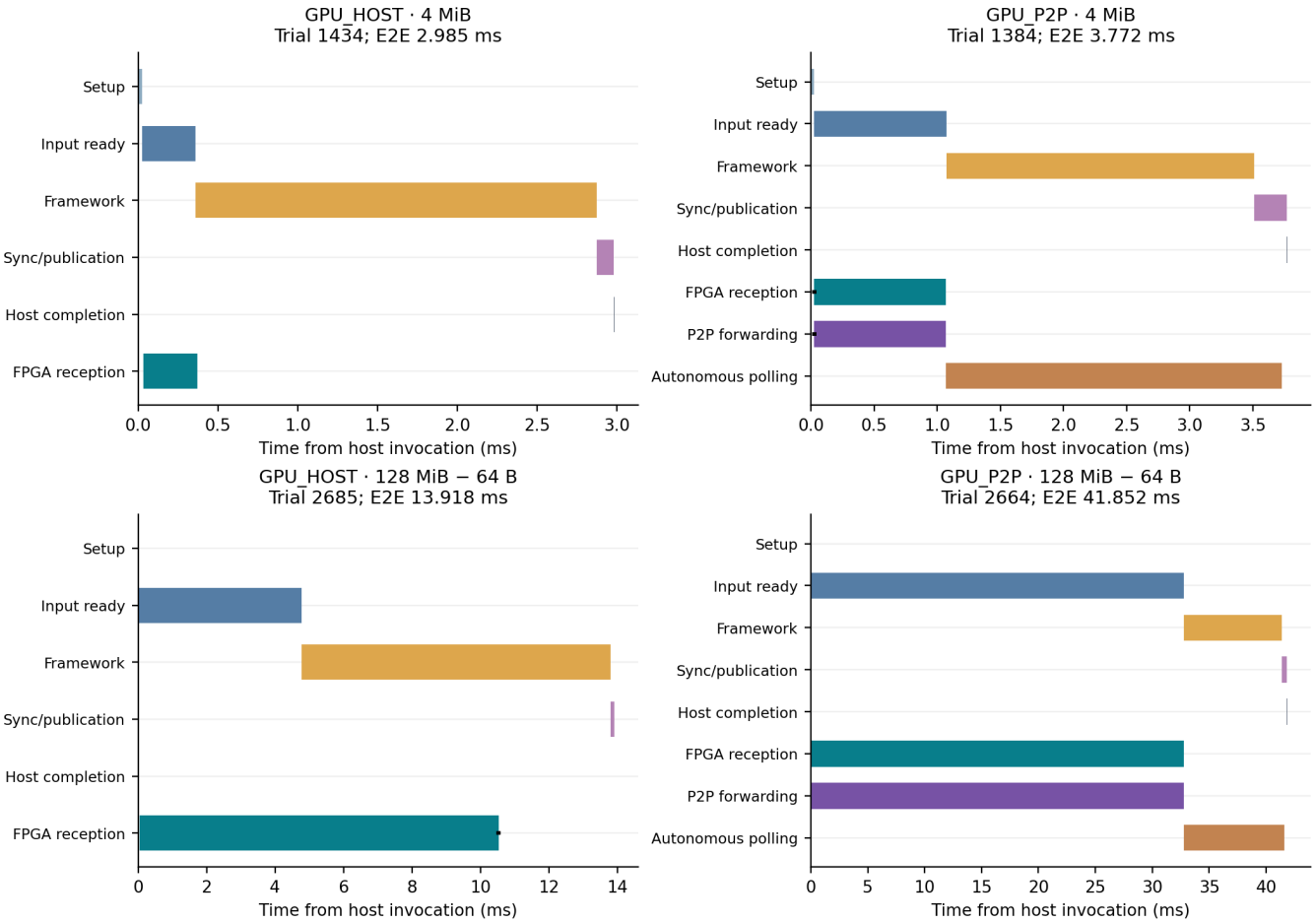
GPU paths: additive elapsed-time breakdowns



Top: resident; bottom: warm-cache file-inclusive. Each stack sums exactly to its mean total. Setup includes mailbox initialization and control preparation. Input-ready wait includes transfer submission/waiting. Framework work ends when result publication begins. Synchronization/publication can include outstanding model work, score conversion, and transport; it is not a pure result-DMA measurement.

Tiny setup, handoff, and completion segments are retained even when visually narrow. Phase means and source references are exported in data/phase_means.csv.

GPU execution timelines and overlap

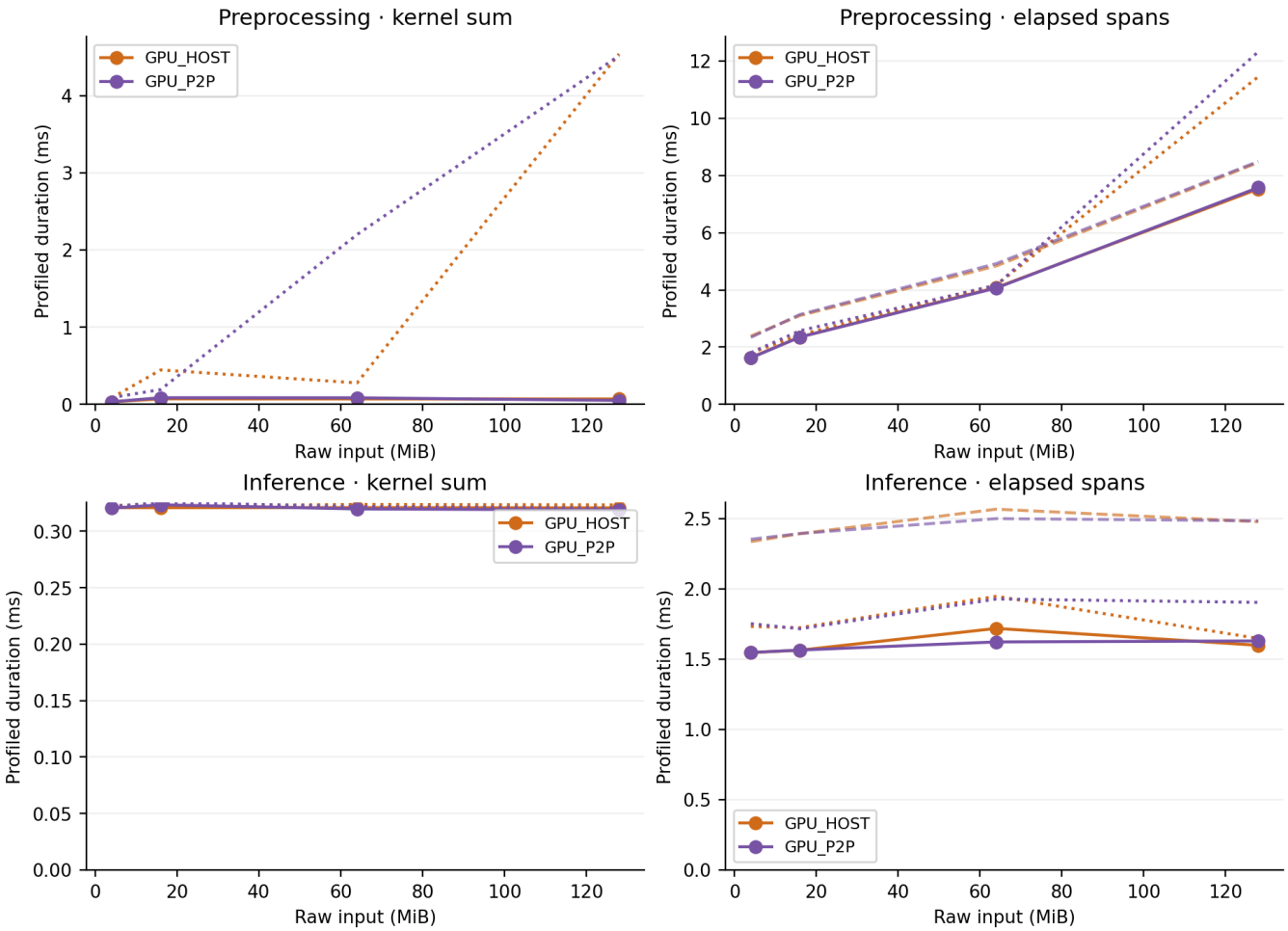


Each panel uses one actual resident trial nearest its workload's median E2E. Host phases and calibrated FPGA intervals share the host-invocation origin. Black endpoint marks show feasible clock-mapping bounds; these are usually smaller than the plotted line width. Exact trial IDs and intervals are exported.

GPU_HOST submits its host-to-GPU copy and FPGA input transfer from the same host call; input-ready means the GPU copy completed, not necessarily that the FPGA accepted its final byte. In P2P, forwarding and reception overlap. FPGA polling overlaps GPU/framework work and publication; adding the polling window to these phases would double-count elapsed time.

The host-to-GPU HIP copy duration is reported separately on page 7: its host-aligned start was not recorded, so no artificial start time is drawn here.

GPU preprocessing and inference diagnostics

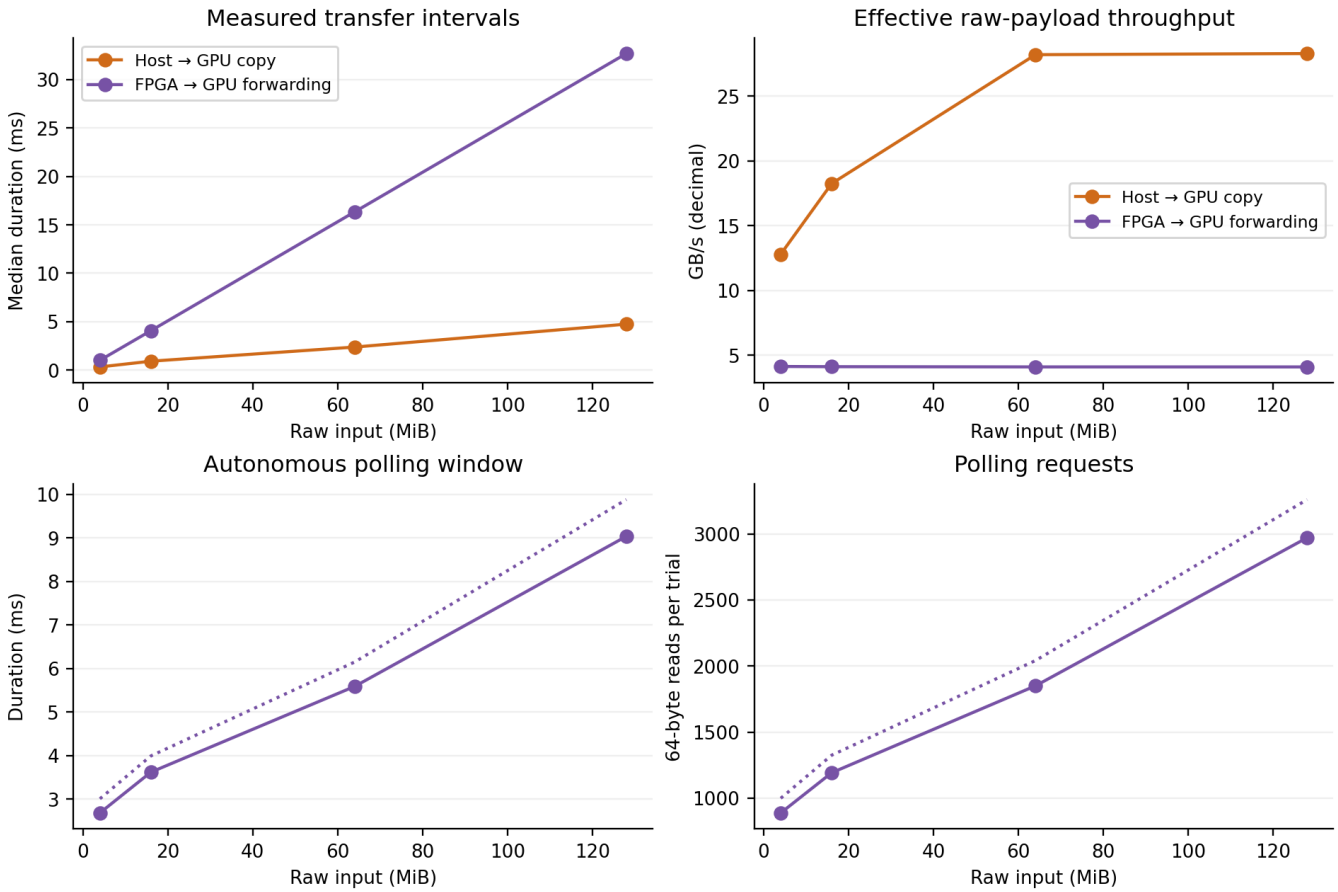


Left: sum of attributed device kernel durations. Right: first-to-last-kernel device span (solid median; dotted p95), with synchronized host-phase median dashed. Twenty diagnostic calls per workload/path were captured separately from primary measurements; both paths have 8 preprocessing kernels and 45 inference kernels where confirmed by the trace tables.

The separation between kernel sums, device spans, and synchronized host spans makes framework scheduling, launch gaps, and synchronization visible. Kernel sums exclude gaps and may not equal exclusive critical-path execution if kernels overlap. A small kernel sum alone does not imply a small system-level inference latency.

These profiles cannot partition an unprofiled E2E bar into exact preprocessing and inference fractions. Profiling phases were explicitly synchronized; first-use effects and profiling overhead can affect their p95 values. No profiled call is included in headline E2E statistics.

P2P transport and autonomous polling

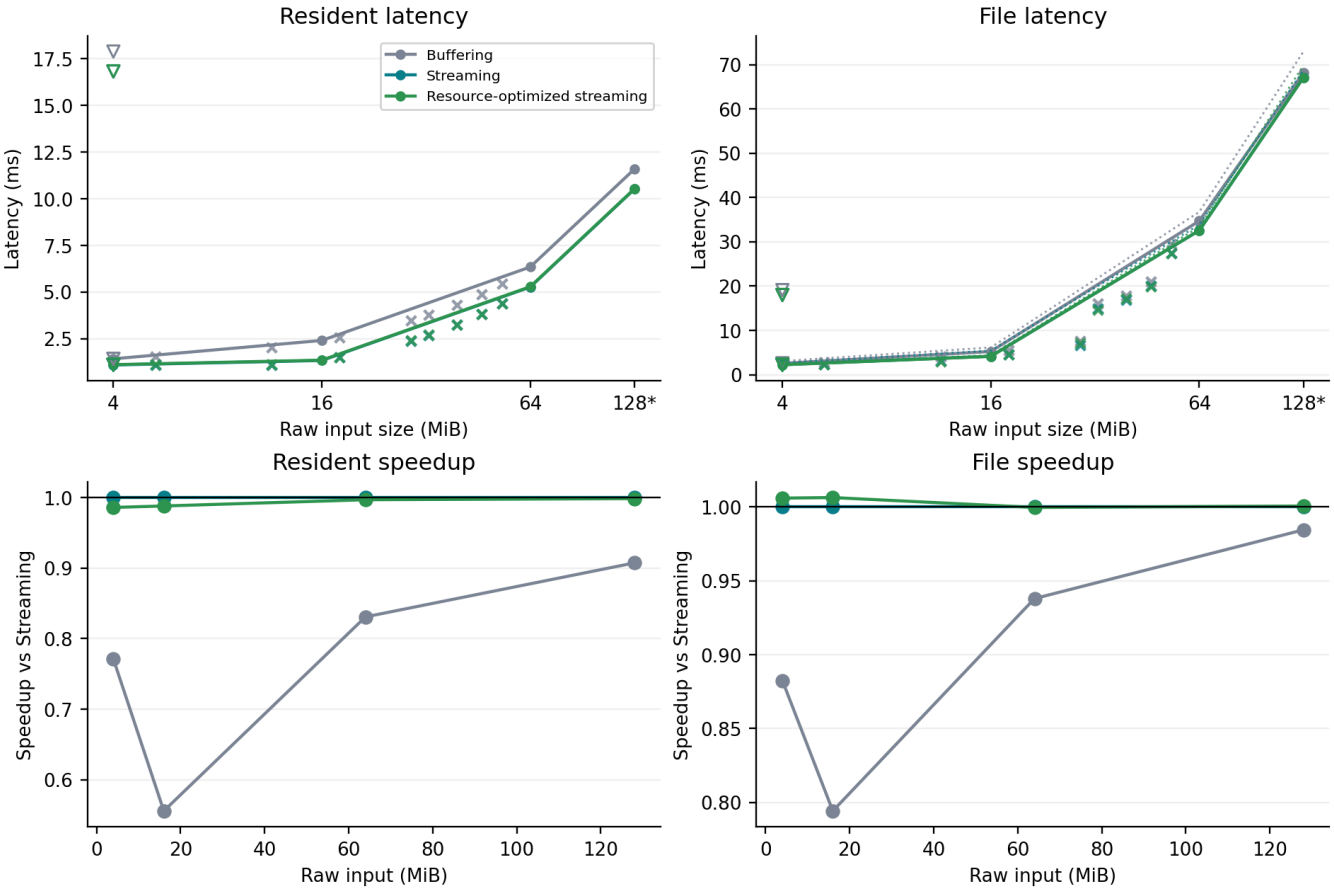


At the largest input, GPU_P2P forwards data in 32.682 ms; the GPU_HOST HIP copy interval is 4.744 ms. Effective bandwidth divides raw payload bytes by the median interval. These are different transport boundaries: P2P includes the FPGA forwarding path; the host-copy interval is measured by HIP events.

The autonomous polling window starts with FPGA polling and ends when the result payload arrives. It includes waiting for the GPU verdict; it is not isolated polling overhead. Each request reads 64 bytes. Request counts and durations describe observed traffic, not GPU compute utilization.

Correctness evidence includes 26 deliberately delayed publications during 520-call P2P admission, repeated autonomous polls, and a quiescent polling engine at completion. No poll-engine error was accepted. Cross-root/NUMA topology is documented, but these measurements do not isolate which hardware mechanism causes the transfer penalty.

Buffering, Streaming, and resource optimization

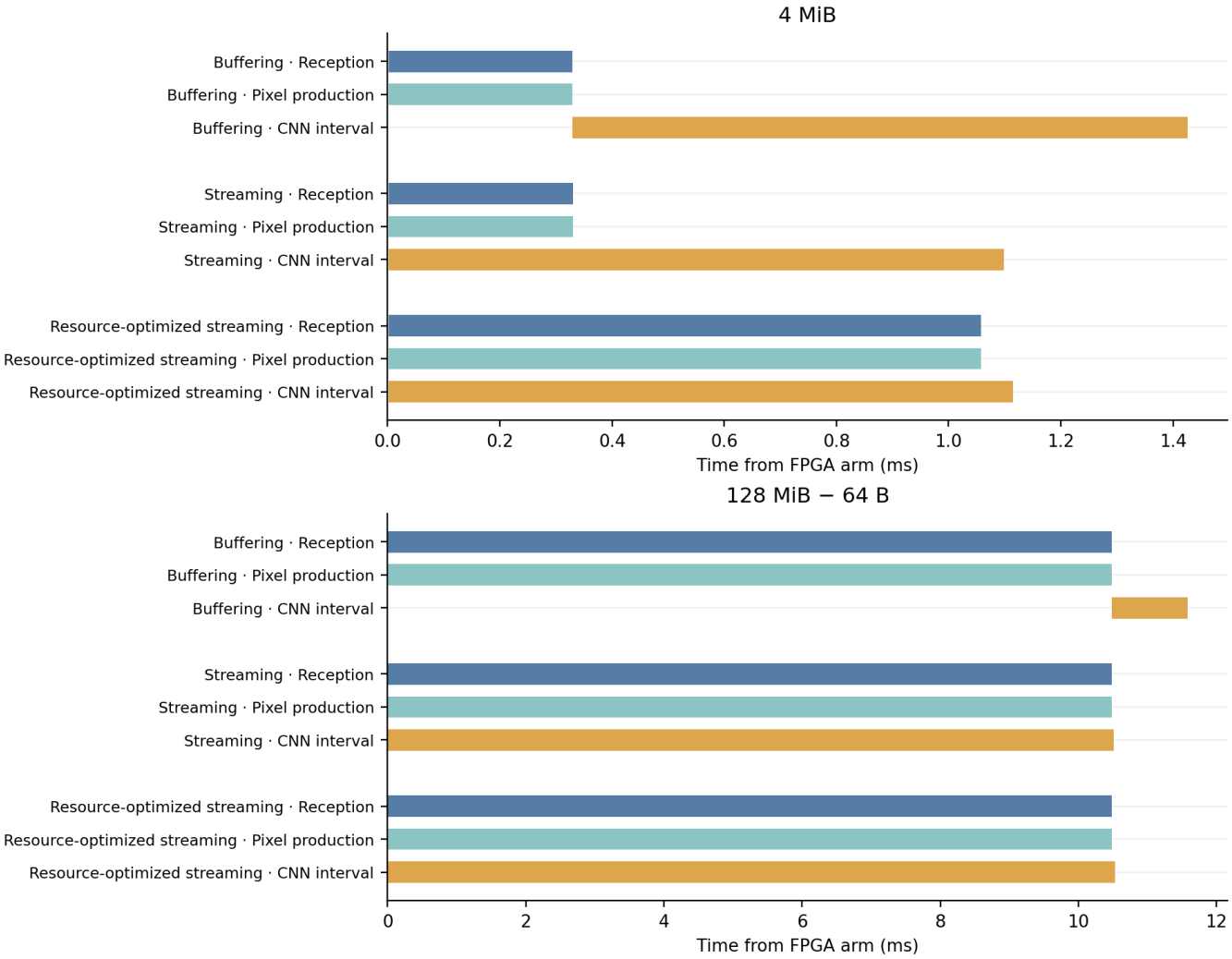


Speedup is Streaming latency divided by the compared variant's latency. Above 1 means faster than Streaming; below 1 means slower. Median speedups are ratios of medians from independent runs, not paired per-trial ratios.

At 4 MiB, Streaming is 1.30× faster than Buffering for resident input. At the largest frame it is 1.10× faster. Resource-optimized streaming adds 15.6 µs at 4 MiB and 15.8 µs at the largest frame.

Small file-inclusive differences should not be interpreted as hardware wins: file read/copy variability can be larger than the FPGA invocation difference. The baseline repeat and rechecks constrain drift but do not remove all host variability.

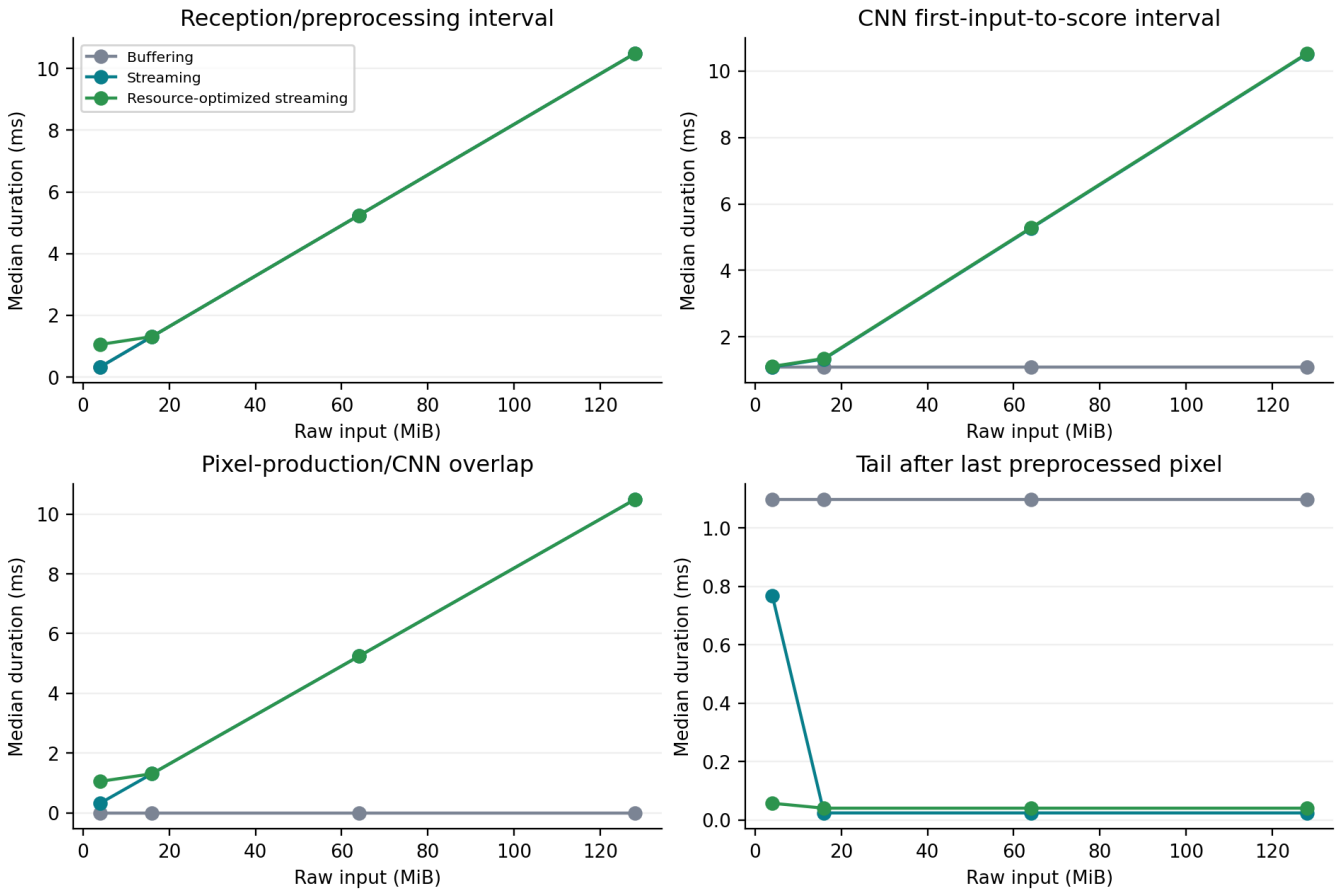
How streaming hides the CNN interval



Timelines use actual resident trials nearest the median total. All lanes on this page use the FPGA clock, relative to arm; no host/FPGA alignment is needed. Pixel production starts at the first output pixel. Reception/preprocessing can begin earlier.

Buffering waits for the complete input and pixel frame before CNN consumption. Streaming begins consumption early, hiding much of the CNN interval behind input arrival. Resource-optimized streaming preserves that overlap. These are observed timelines: the CNN lane includes periods waiting for incoming pixels.

FPGA pipeline intervals across input sizes

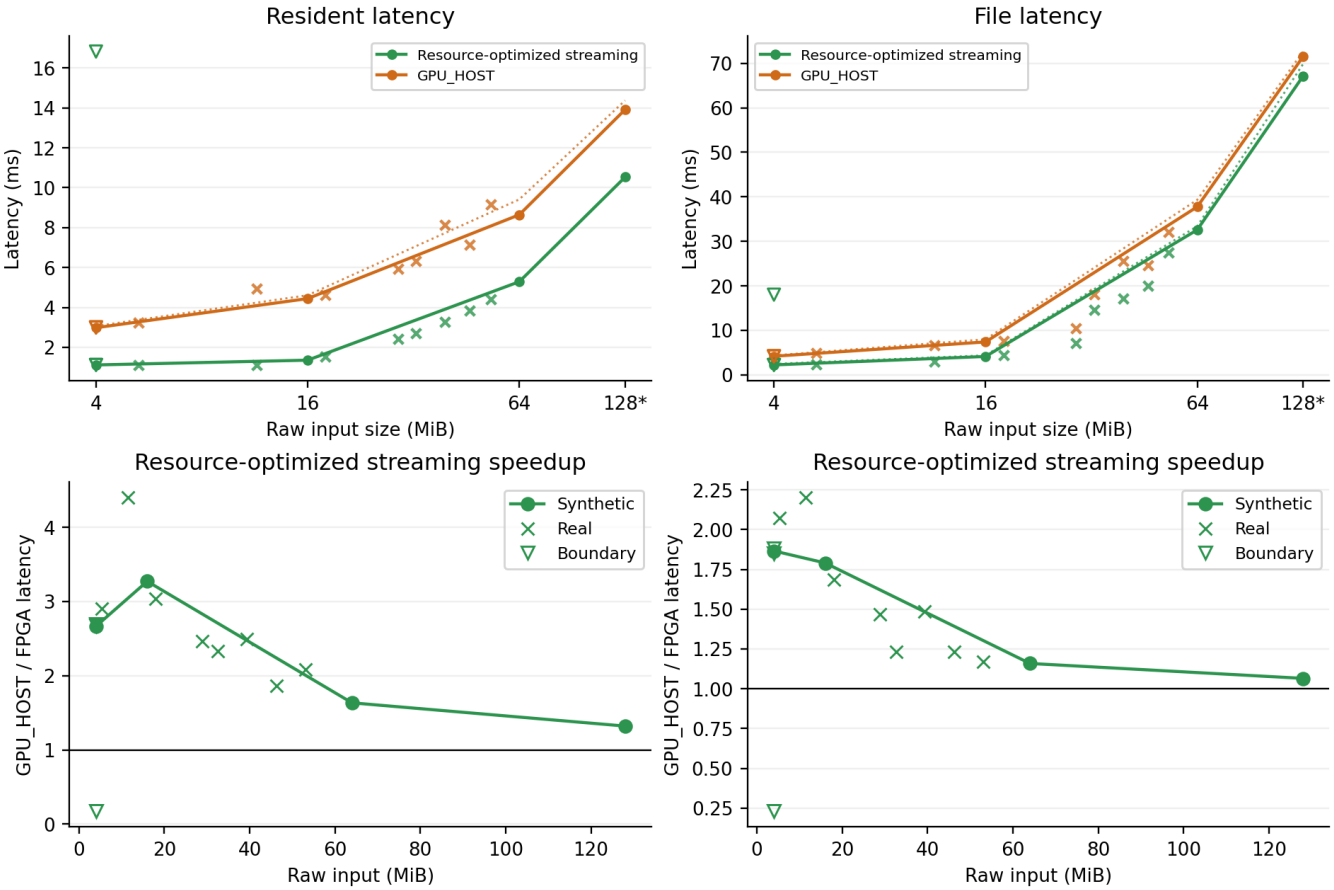


The reception/preprocessing interval runs from first raw input acceptance to last produced pixel. The CNN interval runs from first consumed pixel to score. The overlap metric covers the intersection of pixel production and CNN execution. The tail runs from last produced pixel to FPGA verdict; it is not standalone CNN compute.

At large inputs, a long streaming CNN interval mainly reflects its duration along the arriving stream. The exposed tail is much shorter than the full interval. Buffering retains an approximately fixed post-preprocessing CNN cost. FIFO pressure and input stalls can change observed phase lengths without causing failure.

These four quantities have different boundaries and overlap. They are plotted as separate curves rather than summed into a misleading total.

Resource-optimized streaming versus GPU_HOST



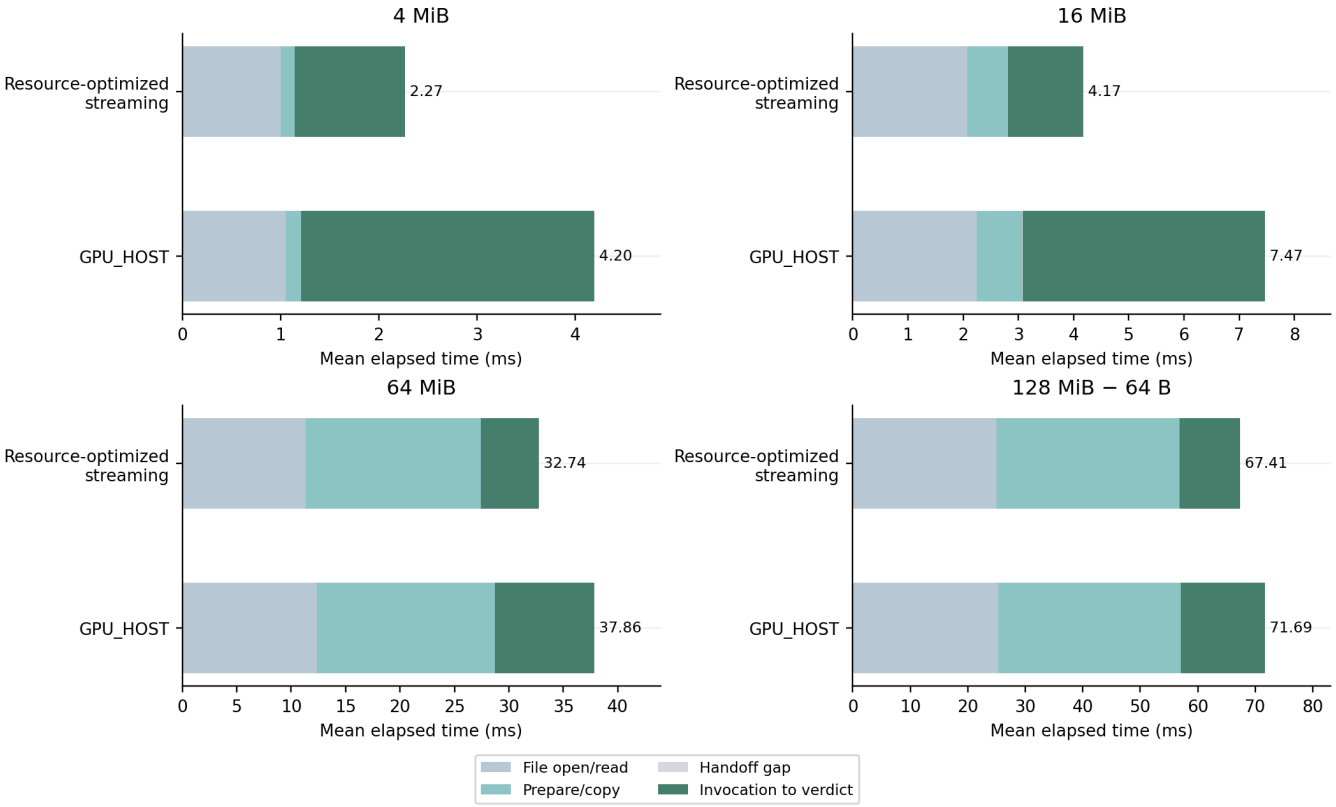
At 4 MiB, Resource-optimized streaming achieves 2.67× resident speedup and 1.87× file-inclusive speedup versus GPU_HOST. At the largest frame, the speedups are 1.32× and 1.07×, respectively.

The boundary triangles include the reversal at 4,194,240 bytes: the resident FPGA speedup is only 0.178× (GPU_HOST is 5.63× faster). The adjacent one-byte-larger case returns to the faster FPGA path. Do not interpolate across this discontinuity.

The GPU_HOST/FPGA latency ratio uses matching input bytes and timing scope. A value above 1 favors Resource-optimized streaming. Solid synthetic curves and real-input crosses show how the advantage changes with workload size; p95 latency remains visible in the upper panels.

The comparison includes the implemented TensorFlow runtime, transfers, and completion path. It does not predict an ideal fused GPU implementation, and it does not establish equal classification accuracy for FP32 and W8A8.

Where the FPGA/GPU_HOST difference is spent

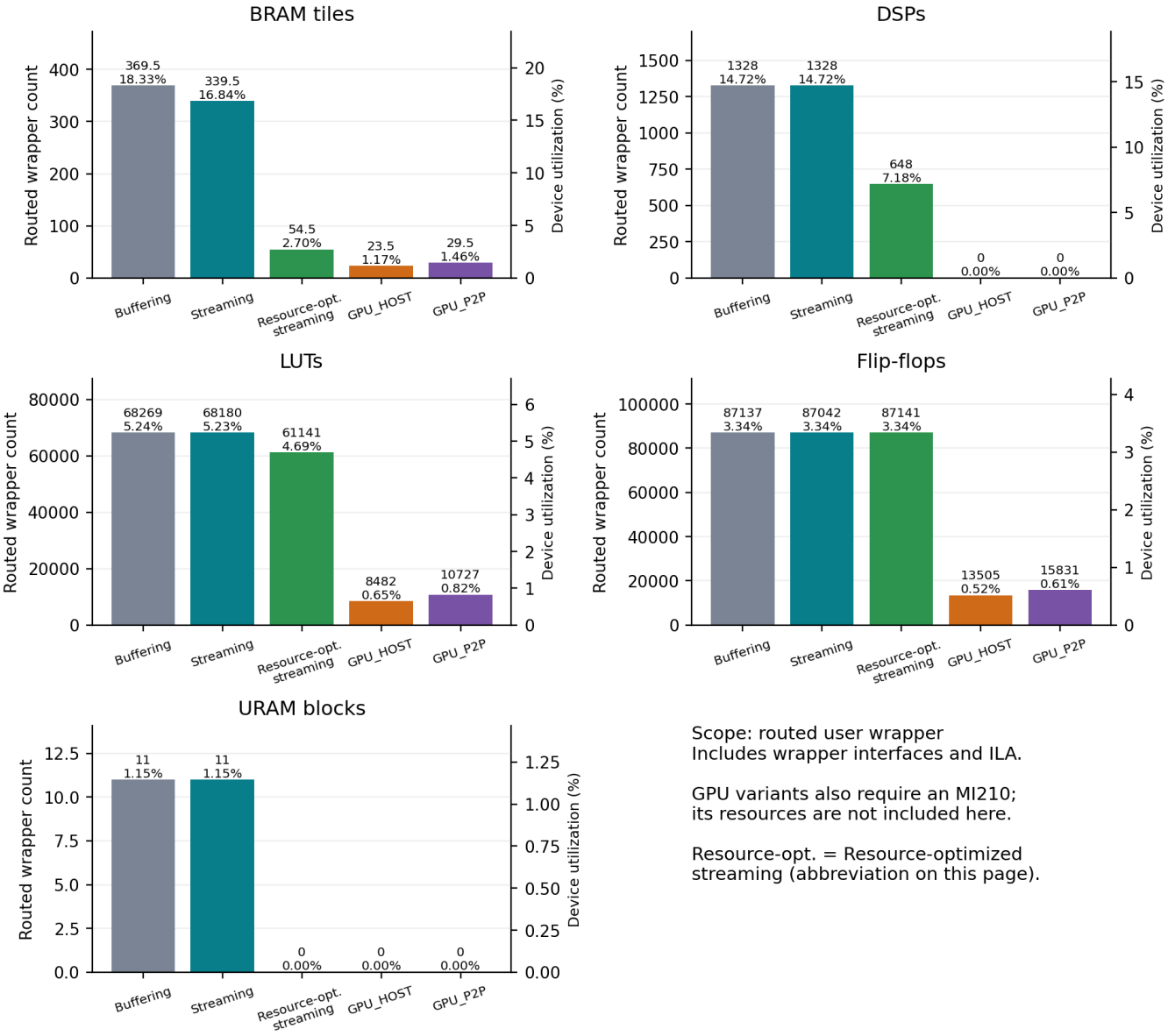


The same additive file-time boundaries are used for both systems: file open/read, host preparation/copy, handoff gap, and invocation to host-observed verdict. Means are plotted so all segments sum exactly. Initial allocation and diagnostic readback remain outside the measured endpoint.

At large sizes, file read and preparation costs dilute the system-level effect of faster inference. At small sizes, the invocation segment contributes more strongly to the FPGA advantage. Apparent differences in file segments are measured host effects; they are not attributable to the FPGA optimization itself.

For the GPU invocation subphases, refer to page 4. For FPGA overlap inside invocation, refer to pages 9–10. Combining their overlapping internal intervals with this outer stack would double-count time.

Resource savings and device utilization



Scope: routed user wrapper
Includes wrapper interfaces and ILA.

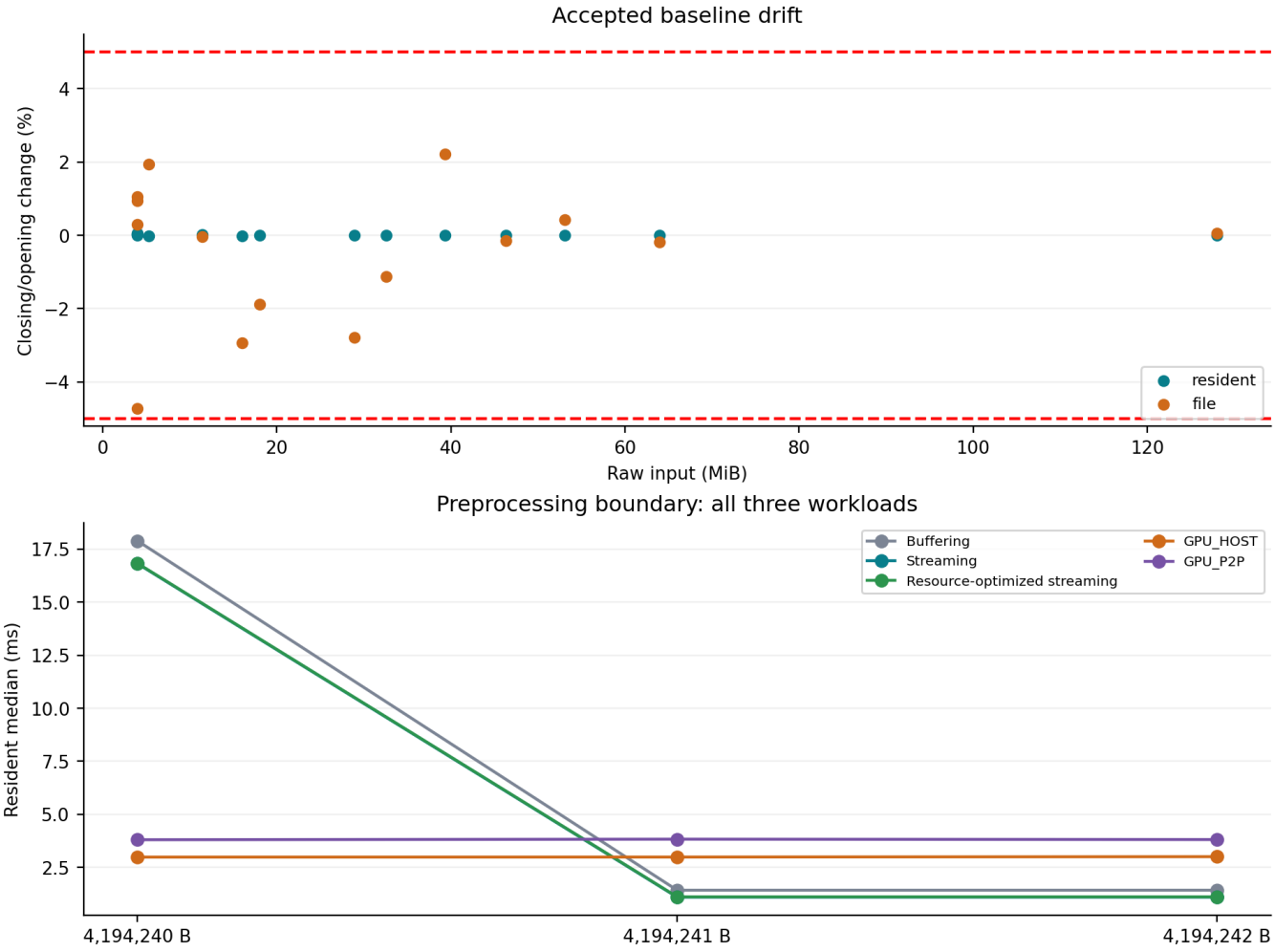
GPU variants also require an MI210;
its resources are not included here.

Resource-opt. = Resource-optimized streaming (abbreviation on this page).

Resource-optimized streaming changes wrapper BRAM from 339.5 to 54.5 tiles (–83.95%), DSPs from 1328 to 648 (–51.20%), and URAM from 11 to 0. BRAM device utilization falls from 16.84% to 2.70%; DSP utilization falls from 14.72% to 7.18%.

Bars and right axes describe the routed user wrapper, including interfaces and ILA. Full-design counts include shell overhead and appear on page 18. GPU-side FPGA resource savings do not account for the MI210's compute, memory, power, or cost.

Validation, drift, and boundary workloads



The initial closing baseline had at most 0.051% resident median drift. Two file-inclusive cases exceeded 5% because read/copy times shifted. A predefined recheck repeated both cases across all variants with the original 128 MiB allocation and a closing baseline; accepted drift was 1.95% (5,540,996 bytes) and 2.22% (41,278,872 bytes). The maximum accepted drift across all cases/scopes is 4.73%.

The boundary panel shows a repeatable latency discontinuity, not a correctness failure: at 4,194,240 bytes, Streaming and Resource-optimized streaming spend about 16.777 ms in preprocessing; at 4,194,241 bytes they return to approximately 1.1 ms resident E2E. GPU paths do not show the same discontinuity. Isolating the implementation cause requires separate investigation.

Only those two file comparisons use rechecks/file_02. All other accepted measurements retain their original cohort. The smaller-buffer file_01 recheck is diagnostic only. No warmup, rejected attempt, or diagnostic profile call enters the main latency distributions.

FPGA board scores match quantized references exactly. GPU scores match CPU FP32 references within 1e-4 absolute/relative tolerance, with exact verdicts. The 520-call per-image admission, timestamp gates, and file hashes passed; these are execution checks, not a new held-out classification accuracy study.

Complete resident-input latency table

Each cell: median / p95 in milliseconds; 200 measured trials after 20 warmups. B = boundary, R = real, S = synthetic. Labels are abbreviated only in this dense table: Resource-opt. streaming = Resource-optimized streaming.

Input bytes / type	Buffering	Streaming	Resource-opt. streaming	GPU_HOST	GPU_P2P
4,194,240 / B	17.880 / 17.882	16.807 / 16.808	16.824 / 16.825	2.990 / 3.067	3.808 / 4.062
4,194,241 / B	1.431 / 1.433	1.103 / 1.105	1.119 / 1.121	2.992 / 3.065	3.834 / 4.051
4,194,242 / B	1.431 / 1.432	1.103 / 1.105	1.119 / 1.121	3.011 / 3.357	3.816 / 4.059
4,194,304 / S	1.431 / 1.432	1.104 / 1.105	1.119 / 1.121	2.985 / 3.067	3.771 / 4.089
5,540,996 / R	1.536 / 1.537	1.104 / 1.105	1.120 / 1.121	3.256 / 3.344	4.232 / 4.487
12,004,440 / R	2.041 / 2.042	1.104 / 1.105	1.119 / 1.121	4.926 / 5.051	7.164 / 7.456
16,777,216 / S	2.414 / 2.416	1.342 / 1.343	1.358 / 1.359	4.444 / 4.612	7.765 / 8.126
18,876,484 / R	2.578 / 2.580	1.506 / 1.507	1.521 / 1.523	4.615 / 4.728	8.487 / 8.775
30,323,588 / R	3.473 / 3.474	2.400 / 2.401	2.416 / 2.417	5.954 / 6.068	12.489 / 12.755
34,162,708 / R	3.772 / 3.773	2.700 / 2.700	2.716 / 2.716	6.342 / 6.457	13.626 / 13.911
41,278,872 / R	4.328 / 4.329	3.255 / 3.257	3.271 / 3.272	8.151 / 8.301	17.007 / 17.391
48,540,364 / R	4.896 / 4.897	3.823 / 3.823	3.839 / 3.840	7.154 / 7.842	17.463 / 18.262
55,676,772 / R	5.453 / 5.454	4.380 / 4.381	4.396 / 4.397	9.170 / 9.903	21.069 / 21.914
67,108,864 / S	6.346 / 6.347	5.273 / 5.274	5.290 / 5.290	8.645 / 9.416	22.054 / 22.659
134,217,664 / S	11.589 / 11.590	10.516 / 10.517	10.532 / 10.533	13.917 / 14.373	41.851 / 42.753

Resident input starts at the host invocation. File-inclusive timing includes warm-cache open/read and preparation. Results are selected by case ID, scope, and accepted evidence path, not by taking the newest attempt blindly.

Complete file-inclusive latency table

Each cell: median / p95 in milliseconds; 200 measured trials after 20 warmups. B = boundary, R = real, S = synthetic. Labels are abbreviated only in this dense table: Resource-opt. streaming = Resource-optimized streaming.

Input bytes / type	Buffering	Streaming	Resource-opt. streaming	GPU_HOST	GPU_P2P
4,194,240 / B	19.107 / 19.293	17.971 / 18.139	18.020 / 18.207	4.182 / 4.411	5.198 / 5.764
4,194,241 / B	2.553 / 2.759	2.216 / 2.340	2.234 / 2.397	4.208 / 4.558	5.152 / 5.700
4,194,242 / B	2.537 / 2.795	2.203 / 2.423	2.234 / 2.367	4.136 / 4.507	5.120 / 5.475
4,194,304 / S	2.556 / 3.071	2.256 / 2.645	2.243 / 2.460	4.185 / 4.372	5.007 / 5.267
5,540,996 / R	2.756 / 2.981	2.304 / 2.473	2.384 / 2.614	4.947 / 5.156	5.870 / 6.133
12,004,440 / R	4.020 / 4.301	3.064 / 3.810	3.028 / 3.175	6.666 / 6.883	9.508 / 10.187
16,777,216 / S	5.238 / 6.140	4.161 / 5.435	4.135 / 4.381	7.403 / 7.984	11.169 / 11.784
18,876,484 / R	5.591 / 5.993	4.522 / 5.020	4.497 / 5.241	7.583 / 7.993	11.703 / 12.354
30,323,588 / R	7.581 / 8.187	6.608 / 7.341	7.116 / 7.707	10.431 / 11.079	16.985 / 17.649
34,162,708 / R	16.122 / 17.064	14.892 / 16.164	14.643 / 15.145	18.067 / 18.846	26.804 / 27.516
41,278,872 / R	17.876 / 18.739	16.825 / 17.429	17.226 / 18.624	25.566 / 27.689	34.370 / 36.622
48,540,364 / R	21.022 / 22.569	20.058 / 20.707	19.984 / 20.285	24.579 / 25.722	35.171 / 36.555
55,676,772 / R	29.462 / 31.346	27.385 / 27.838	27.413 / 29.571	32.084 / 34.324	40.626 / 41.383
67,108,864 / S	34.723 / 36.756	32.569 / 33.860	32.582 / 33.382	37.758 / 39.322	52.154 / 53.796
134,217,664 / S	68.211 / 72.984	67.150 / 68.785	67.113 / 69.835	71.486 / 72.638	100.735 / 103.400

Resident input starts at the host invocation. File-inclusive timing includes warm-cache open/read and preparation. Results are selected by case ID, scope, and accepted evidence path, not by taking the newest attempt blindly.

Recheck cohort: the 5,540,996-byte and 41,278,872-byte rows use the capacity-matched file_02 recheck for all variants. Original resident measurements remain unchanged. Cross-row trends may include cohort effects; same-row variant comparisons remain matched.

GPU profiling detail and timestamp definitions

Profile table: 4 MiB and largest frame, median / p95 (ms), 20 synchronized diagnostic trials per row. Full all-workload profile statistics are exported in data/gpu_profiles.json.

Path / size / phase	Kernel sum	Device span	Host synchronized
GPU_HOST / 4 MiB / preprocessing	0.033 / 0.084	1.639 / 1.798	2.386 / 2.565
GPU_HOST / 4 MiB / inference	0.321 / 0.323	1.547 / 1.732	2.336 / 2.516
GPU_HOST / 128 MiB – 64 B / preprocessing	0.070 / 4.539	7.527 / 11.450	8.452 / 12.331
GPU_HOST / 128 MiB – 64 B / inference	0.321 / 0.323	1.598 / 1.647	2.478 / 2.788
GPU_P2P / 4 MiB / preprocessing	0.034 / 0.088	1.623 / 1.809	2.342 / 2.533
GPU_P2P / 4 MiB / inference	0.321 / 0.322	1.547 / 1.753	2.353 / 2.575
GPU_P2P / 128 MiB – 64 B / preprocessing	0.051 / 4.516	7.577 / 12.320	8.489 / 13.349
GPU_P2P / 128 MiB – 64 B / inference	0.319 / 0.321	1.630 / 1.904	2.485 / 2.969

Quantity	Boundary / meaning
File E2E	File open/read start → host-observed verdict
Resident E2E	Host invocation start → host-observed verdict
Host→GPU copy	HIP copy_begin → copy_end; device event duration, no aligned host start
P2P forwarding	FPGA CSR18 → CSR19; first/last forwarded payload timing
Polling window	CSR38 → CSR41; polling begins → result payload arrival
Ready→payload	CSR40 → CSR41; readiness response → payload arrival
FPGA preprocessing	First raw acceptance (CSR8) → last pixel produced (CSR11)
FPGA CNN interval	First CNN input (CSR12) → score (CSR14); includes stalls
GPU device span	First attributed kernel start → last attributed kernel end
Kernel sum	Sum of attributed kernel durations; excludes gaps
Publication host phase	Before score conversion/DLPack → completion of publication call

FPGA timestamps are calibrated against 32 MMIO brackets before and after each block. GPU collectors use at most 20 trials per calibration block. GPU XPlane events are correlated with Trace markers and step_num in the trace time domain, not Python wall-clock timestamps.

Resources, evidence, and reproduction

Full-design routed utilization, including shell overhead. Each cell shows count / percentage of U55C capacity. GPU variants still require the external MI210.

Variant	BRAM	DSP	LUT	FF	URAM
Buffering	511.5 / 25.37%	1328 / 14.72%	189158 / 14.51%	259187 / 9.94%	11 / 1.15%
Streaming	481.5 / 23.88%	1328 / 14.72%	188993 / 14.50%	258992 / 9.93%	11 / 1.15%
Resource-optimized streaming	196.5 / 9.75%	648 / 7.18%	181886 / 13.95%	259143 / 9.94%	0 / 0.00%
GPU_HOST	165.5 / 8.21%	0 / 0.00%	125538 / 9.63%	176994 / 6.79%	0 / 0.00%
GPU_P2P	171.5 / 8.51%	0 / 0.00%	131566 / 10.09%	187929 / 7.21%	0 / 0.00%

Campaign: hls4ml/optimize/runs/final_benchmarks_gpu_u55c_20261019. Primary sources: summary.json, measurement_manifest.json, campaign.json, environment_manifest.json, and per-attempt observations/derived/calibration files. Bitstreams and runtimes remain pinned to the campaign identities.

Verification: all 207 measurement-manifest hashes checked; 180 accepted case/scope/iteration cells each contain 200 non-warmup trials, including the closing baseline. Every stored median/p95 is reconciled against raw derived observations. Additive intervals are nonnegative and match per-trial elapsed time. Timestamp gates were enforced during collection; timeline calibration is reconstructed for selected GPU trials.

Report exports: figures/ contains SVG, PNG, and PDF plots. data/statistics.csv contains means, medians, p95, cohort, and evidence pointers. data/phase_means.csv contains additive breakdown values; data/timelines.json records selected trials and boundaries. data/gpu_profiles.json preserves all diagnostic profile summaries. evidence_manifest.json records source and generated-file hashes.

Reproduce: from the repository root, run ml_baseline/.venv/bin/python hls4ml/optimize/reports/optimizations/final_benchmarks_20261019/build_report.py. Uses the existing NumPy/Matplotlib environment and system ReportLab. Reads campaign data and writes only this report directory; no hardware access, rebuild, training, or benchmark replay.

Limits: fixed variant order, one host session, one GPU and FPGA pairing, warm filesystem cache, and different inference precision. File rechecks address observed drift without proving complete stationarity. Resource totals are synthesis/route evidence; latency values are board measurements. No power, energy, cold-storage, concurrent-load, or accuracy-equivalence claims are made.

Reading conclusion: Resource-optimized streaming retains nearly the Streaming latency while substantially reducing wrapper memory and DSPs. GPU_HOST is faster than GPU_P2P here. FPGA paths lead on the real and synthetic workloads, while the slow boundary case reverses that result. The evidence separates transport, observable host phases, overlapping hardware intervals, and diagnostic GPU execution.